Data Science Indonesia Banking Stock Forecasting Platform

End-to-End Machine Learning Project

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Live Demo

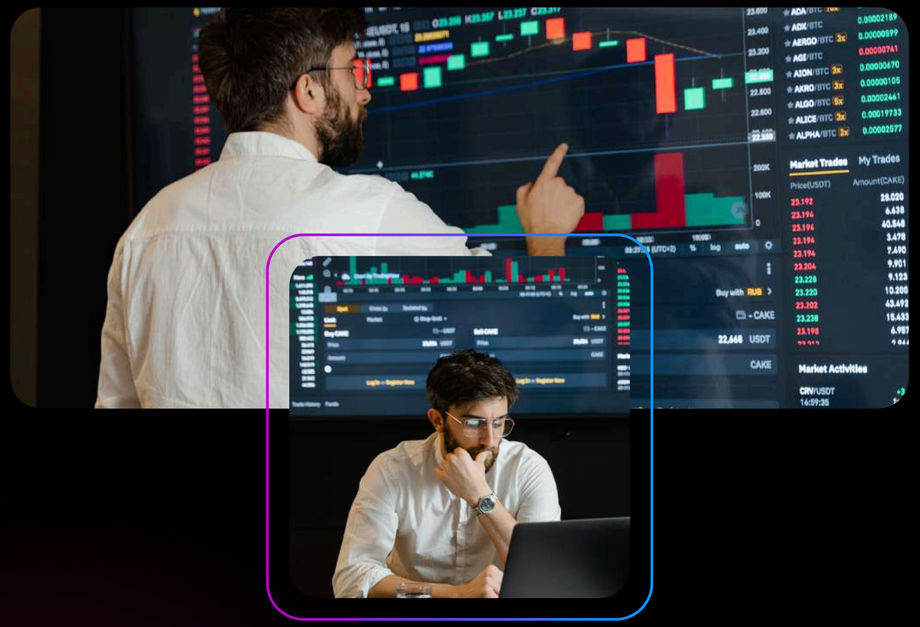
2026 Portofolio Project

https://adjiehf231-indonesia-banking-stock-forecasting-platform.streamlit.app/

End-to-End Machine Learning & Deep Learning System

An enterprise-style web application that forecasts stock prices for Indonesia's top 10 banks using advanced Machine Learning and Deep Learning pipelines.

Designed as a production-ready data science portfolio demonstrating model comparison, time-series engineering, and interactive analytics.



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Learn More

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🎯 Executive Summary

This project delivers a full ML lifecycle implementation for financial time-series forecasting, including:

* Automated data ingestion from Yahoo Finance
* Advanced technical feature engineering
* Multi-model training pipeline
* Automatic best-model selection per ticker
* Interactive Streamlit dashboard
* Reproducible and modular architecture

The system evaluates **Random Forest** , **XGBoost** , and **LSTM** models and dynamically selects the best performer for each bank based on evaluation metrics.

Proyek ini adalah aplikasi prediksi harga saham untuk 10 bank terbesar di Indonesia menggunakan berbagai teknik Machine Learning dan Deep Learning. Aplikasi ini mengambil data historis dari Yahoo Finance dan menghitung berbagai indikator teknis untuk membantu memprediksi pergerakan harga saham di masa depan.

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🧠 Business Problem

Forecasting stock prices in emerging markets like Indonesia presents challenges:

* High volatility
* Non-stationary time series
* Market noise
* Sector-specific dynamics

This project explores how classical ML and deep learning models perform on Indonesian banking equities and builds a scalable experimentation platform.

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🏦 Coverage Universe

Ticker	Bank	Ticker	Bank
BMRI	Bank Mandiri	BRIS	Bank Syariah Indonesia
BBRI	Bank Rakyat Indonesia	BNGA	Bank CIMB Niaga
BBCA	Bank Central Asia	NISP	Bank OCBC Indonesia
BBNI	Bank Negara Indonesia	BNLI	Bank Permata
BBTN	Bank Tabungan Negara	BDMN	Bank Danamon

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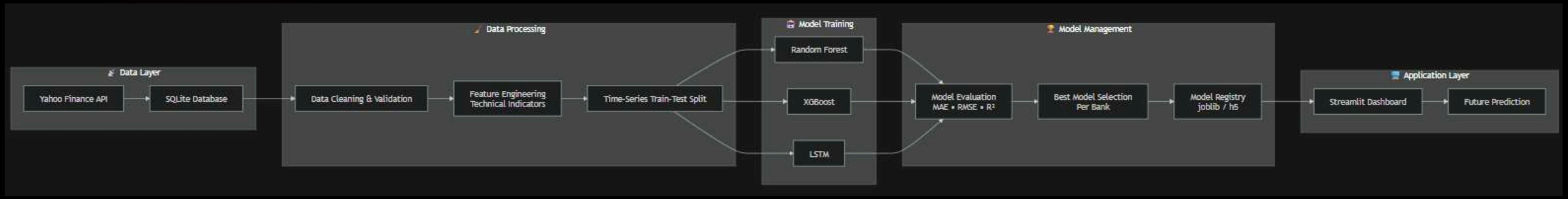
System Architectur



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Key Design Principles

- Modular API layer
- Reproducible pipelines
- Model-agnostic framework
- Scalable for new tickers
- Production-oriented structure

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⚙ Technology Stack

Application Layer

- Streamlit — Interactive analytics UI
- Plotly — High-performance financial charts

Data & Processing

- Pandas / NumPy — Data manipulation
- yfinance — Market data ingestion
- SQLite — Lightweight persistence

Machine Learning

- Scikit-learn — Random Forest
- XGBoost — Gradient boosting
- TensorFlow / Keras — LSTM networks
- Joblib — Model serialization

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Modeling Strategy

Implemented Models

Random Forest

- Robust baseline
- Handles nonlinear tabular patterns
- Low tuning sensitivity

XGBoost

- High-performance gradient boosting
- Strong tabular dominance
- Feature importance support

LSTM

- Sequential deep learning model
- Captures temporal dependencies
- Suitable for long time series

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📊 Feature Engineering

Trend Features

- MA (5, 10, 20, 50, 200)
- EMA (12, 26)

Momentum Features

- RSI
- MACD
- Stochastic Oscillator
- ROC

Volatility Features

- Bollinger Bands
- ATR
- Historical Volatility

Volume Features

- VWAP

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🧪 Model Evaluation Framework

MAE — Mean Absolute Error

RMSE — Root Mean Squared Error

MSE — Mean Squared Error

R^2 Score — Goodness of fit

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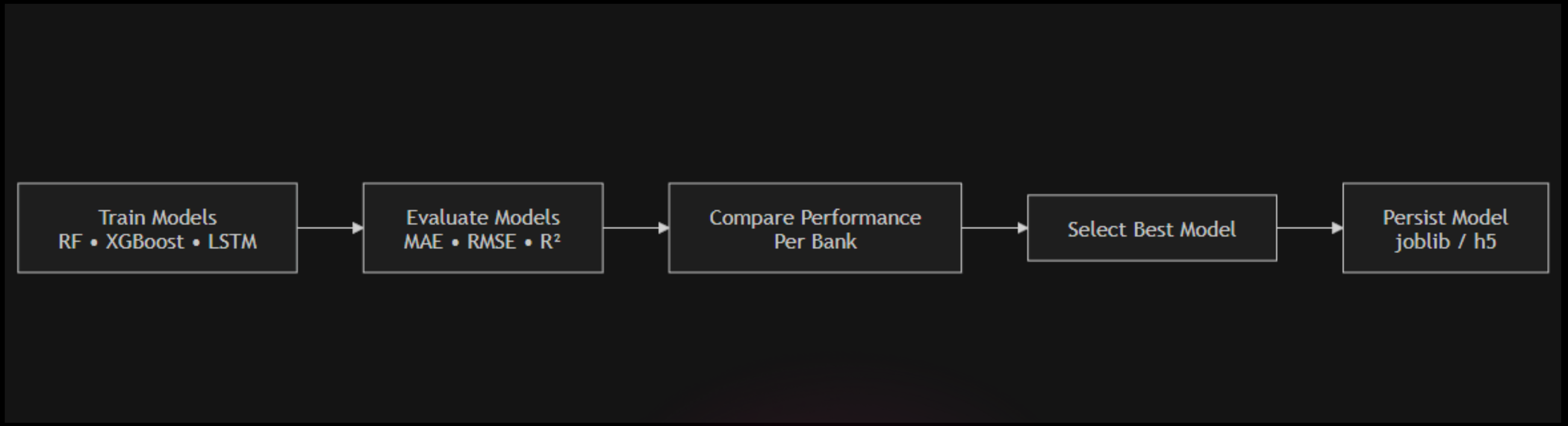
🏆 Automatic Model Selection



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🖥️ Application Features

📊 Interactive Market Dashboard
- Candlestick charts
- Technical overlays
- Multi-bank comparison

🤖 Multi-Model Training
- Automated training pipeline
- Per-ticker evaluation
- Model persistence

🔮 Forecasting Engine
- Multi-horizon prediction
- Historical vs predicted visualization
- Best-model routing

📈 Analytics & Insights
- Dataset diagnostics
- Model metrics
- Descriptive statistics

🧩 Engineering Highlights

- End-to-end ML pipeline
- Modular API architecture
- Production-style project layout
- Model benchmarking framework
- Interactive financial visualization
- Scalable to additional assets

⚠️ Risk Disclaimer

This project is for educational and research purposes only.

- Not financial advice
- Markets are inherently risky
- Past performance ≠ future results
- Always conduct independent research
- Consult licensed financial professionals

🌟 Future Improvements (Roadmap)

- Walk-forward validation
- Hyperparameter optimization
- Transformer-based models
- Live market streaming
- Portfolio backtesting
- Cloud deployment pipeline

👨‍💻 Author
Adjie Hari Fajar

Data Science | Machine Learning | Data Analyst

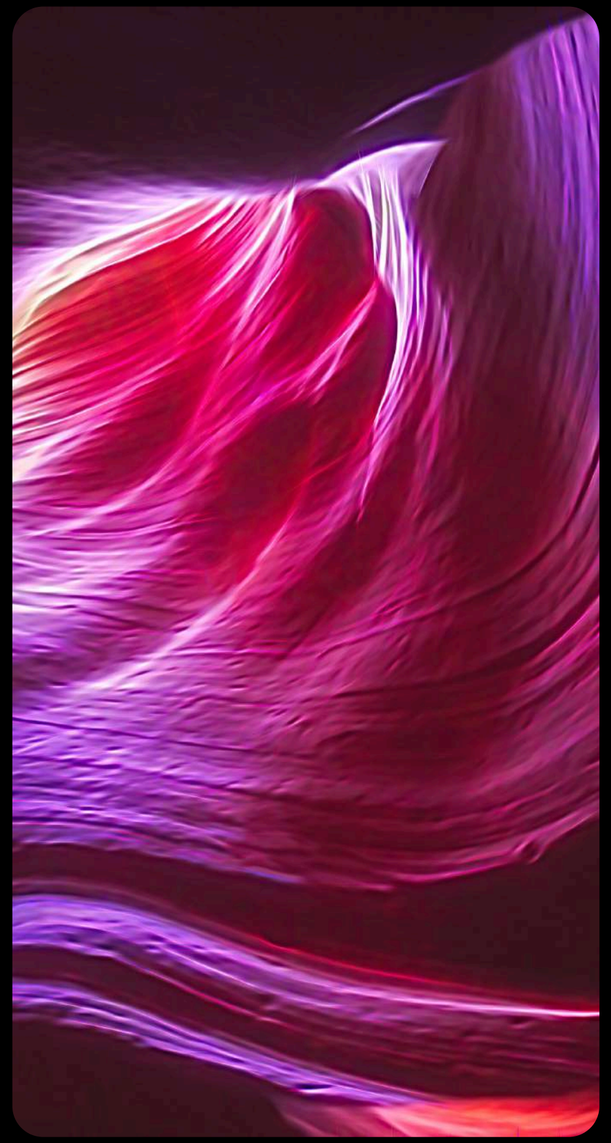
Linkedin
https://www.linkedin.com/in/adjie-hari-fajar-1aa0a629a/

Website
https://fortofolio-ahf231.vercel.app/

Github
https://github.com/adjiehf231/

Email
adjieharifajar2301@gmail.com

2026 Presentation Project



Thank You

FOR YOUR ATTENTION

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